DSA06 DATA HANDLING AND VISUALIZATION

Activity: Identify Seasonality Patterns | Slot C

Tourist Arrivals — Monthly Seasonality Analysis

R. No.: 22 Reg. No.: 192425257 Name: MOTHUKURI MARUTHI

PART A — ORIGINAL ACTIVITY

Scenario

You are the analyst for a hill-station tourism office. The series arrivals (tourist arrivals in thousands) is recorded monthly. Management believes it follows a repeating pattern and wants it identified before planning. The cycle length is 12 observations. Run the supplied R code to create the series x and answer the tasks.

Evidence: Run this code in R (do not edit it)

```
set.seed(32187)
m <- 12                    # observations per cycle (monthly data)
n <- m * 5                 # 5 full cycles
t <- 1:n
trend <- 82.7 + 0.704 * t
arrivals <- round(trend + 16.4 * cos(2 * pi * (t - 4) / m) + rnorm(n, 0, 5.3), 1)
x <- ts(arrivals, frequency = m, start = c(2018, 1))
```

Tasks

- 1. Plot x. State n, the cycle length, the number of full cycles, and whether the trend is rising, falling, or flat. Fit lm(arrivals ~ t) and report the slope per month. [3]
- 2. Use decompose(x) and extract sf (the 12 seasonal values). Name the peak month and trough month with their values, and calculate the seasonal swing (peak minus trough). [4]
- 3. Draw a seasonal plot with one line per cycle against position in the cycle. In how many of the 5 cycles is the highest value in the peak month from Task 2? Is the pattern consistent? [3]
- 4. Detrend with d <- resid(lm(arrivals ~ t)) and compute acf(d, lag.max = 12, plot = FALSE). Report autocorrelation at lag 12 and lag 6, compare with 2/sqrt(n), and interpret the values. [4]
- 5. Calculate the range of arrivals in the first and last cycles and their ratio (last / first). Decide additive or multiplicative seasonality and justify. Decompose using the chosen type, remove the seasonal part, fit lm(adjusted ~ t), and report the slope per month. Give one sentence of advice to management. [6]

Reminder: seasonality repeats at a fixed, known period; a trend is not a season; if the swings grow with the level, the season is multiplicative.

Code of Conduct: I certify that this work is my own. Signature: _____

PART B — WORKED ANSWERS AND R CODE

Run the supplied data-generation code first. The code below prints exact values for seasonal factors, autocorrelations, and fitted slopes. This avoids reporting invented numerical results.

1. Plot, observations, cycles, and trend

```
# Number of observations and cycles
length(arrivals)          # 60 observations
m                         # 12 observations per cycle
length(arrivals) / m      # 5 full cycles

plot(x, type = "o", pch = 16,
     main = "Monthly Tourist Arrivals",
     xlab = "Year", ylab = "Tourist arrivals (thousands)")

trend_fit <- lm(arrivals ~ t)
coef(trend_fit)["t"]      # fitted slope per month
abline(trend_fit, col = "red", lwd = 2)
```

- Expected structure: n = 60 monthly observations, a 12-month cycle, and 5 full cycles.
- The data-generating trend rises by 0.704 thousand tourists per month. Report the fitted slope from lm(arrivals ~ t); random noise can make it slightly different.

2. Seasonal factors, peak, trough, and swing

```
dc <- decompose(x, type = "additive")
sf <- dc$seasonal[1:m]    # 12 seasonal factors
seasonal_table <- data.frame(
  month = month.abb,
  seasonal_factor = as.numeric(sf)
)
seasonal_table

peak_month <- which.max(sf)
trough_month <- which.min(sf)
peak_value <- max(sf)
trough_value <- min(sf)
seasonal_swing <- peak_value - trough_value

month.abb[peak_month]
peak_value
month.abb[trough_month]
trough_value
seasonal_swing
```

- Use the output table to report the peak and trough months and their seasonal factors. The swing is the peak factor minus the trough factor.
- The generated cosine component places the seasonal high point around the fourth position of the cycle, but use the decomposition output for the exact estimated month.

3. Seasonal plot and consistency

```
cycle_no <- rep(1:5, each = 12)
position <- rep(1:12, times = 5)
seasonal_df <- data.frame(
  arrivals = arrivals,
  t = t,
  cycle = factor(cycle_no),
  position = position
```

```
)

library(ggplot2)
ggplot(seasonal_df, aes(position, arrivals,
                        group = cycle, colour = cycle)) +
  geom_line() + geom_point(size = 1.5) +
  scale_x_continuous(breaks = 1:12, labels = month.abb) +
  theme_minimal() +
  labs(title = "Tourist Arrivals by Yearly Cycle",
       x = "Month within cycle", y = "Arrivals (thousands)",
       colour = "Cycle")

# Find the position of the maximum in each year
peak_positions <- sapply(1:5, function(k) {
  vals <- arrivals[((k-1)*12+1):(k*12)]
  which.max(vals)
})
peak_positions
sum(peak_positions == peak_month)
```

- The final count gives the number of years whose highest observed value occurs in the estimated peak month. If some maxima differ, the repeating pattern can still be broadly consistent because random noise affects individual months.

4. Autocorrelation after detrending

```
d <- resid(lm(arrivals ~ t))
ac <- acf(d, lag.max = 12, plot = FALSE)
acf_values <- as.numeric(ac$acf)
acf_lag6 <- acf_values[6 + 1]
acf_lag12 <- acf_values[12 + 1]
bound <- 2 / sqrt(length(d))

acf_lag12
acf_lag6
bound
```

- The ACF array includes lag 0 at index 1; therefore lag 6 is index 7 and lag 12 is index 13.
- Compare the absolute autocorrelations with 2/sqrt(n). A lag-12 value outside the approximate bound supports annual seasonality in monthly data. Lag 6 may be negative because it is half of the 12-month cycle.

5. Additive or multiplicative seasonality and adjusted trend

```
first_cycle <- arrivals[1:12]
last_cycle <- arrivals[(length(arrivals)-11):length(arrivals)]
range_first <- max(first_cycle) - min(first_cycle)
range_last <- max(last_cycle) - min(last_cycle)
range_ratio <- range_last / range_first

range_first
range_last
range_ratio

# The generated seasonal amplitude is roughly constant,
# so additive decomposition is a reasonable choice.
dc_add <- decompose(x, type = "additive")
adjusted <- x - dc_add$seasonal
adjusted_fit <- lm(as.numeric(adjusted) ~ t)
coef(adjusted_fit)["t"]
```

- If the first and last ranges are similar and their ratio is near 1, the data are more consistent with additive seasonality. This ratio is only a rough diagnostic because noise affects both ranges.
- The seasonal component is generated with a roughly fixed amplitude, so additive decomposition is the appropriate starting point.
- Management advice: use the annual seasonal pattern to plan staffing, accommodation, transport, and promotions, while accounting for the upward trend in tourist arrivals.

Final summary

- The series contains 60 monthly observations, representing five complete 12-month cycles.
- The fitted linear model estimates the overall increase in tourist arrivals per month.
- Decomposition estimates 12 seasonal factors and identifies peak and trough months.
- The seasonal plot and ACF at lag 12 help assess annual repetition.
- The generated seasonal swings have approximately constant amplitude, supporting an additive model.
- Tourism planning should combine seasonal peaks with the upward trend and uncertainty in monthly demand.